

EMCH 260

SOLID MECHANICS

Ming Hu

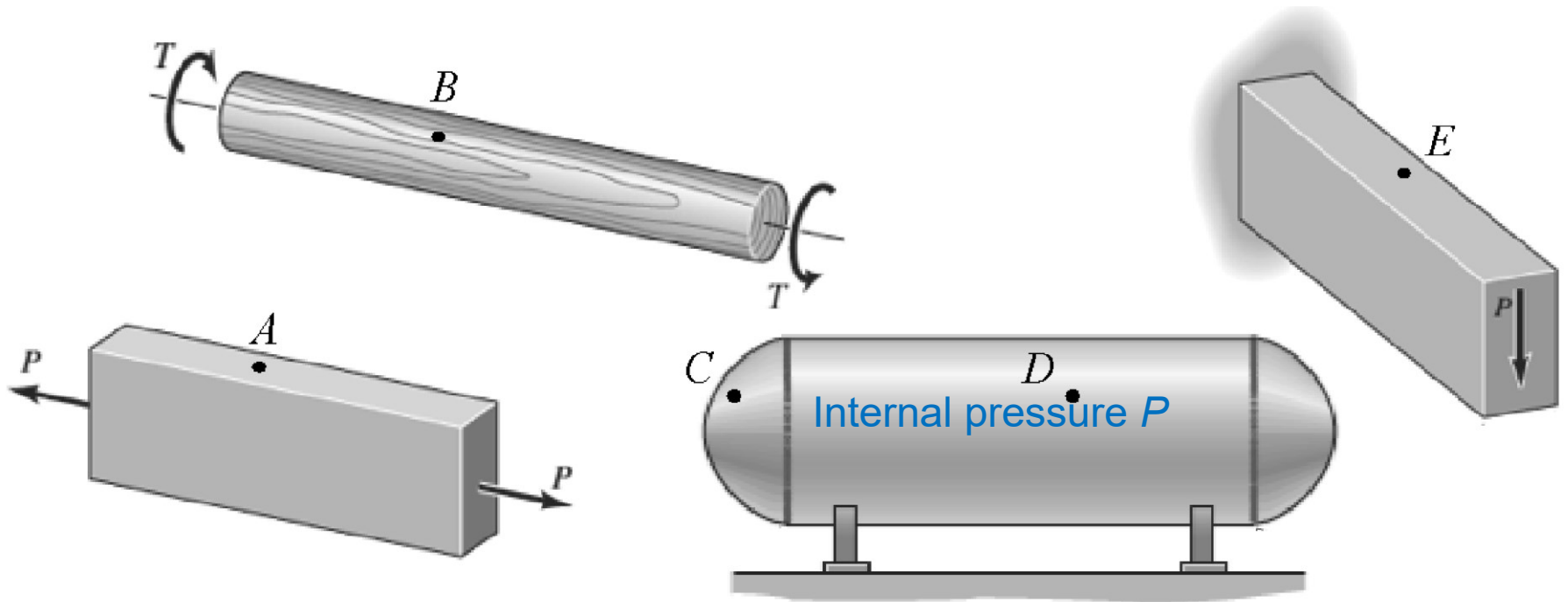
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Stress Transformation

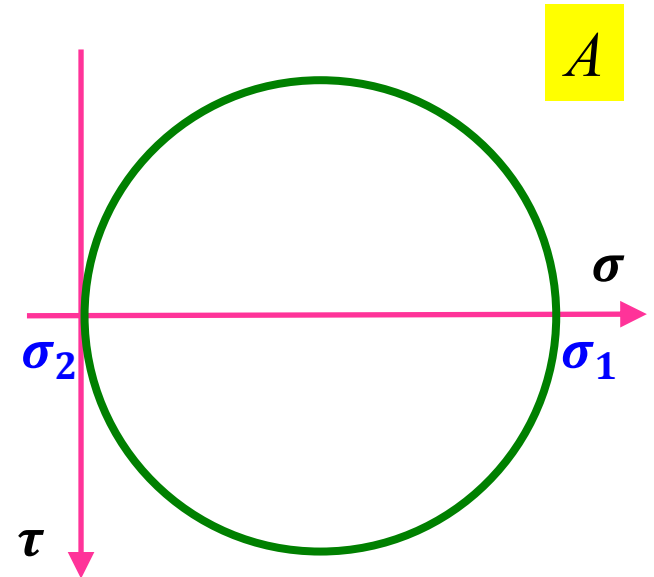
Chapter 9.4

- ✓ Know how to use the Mohr's circles to solve complex problems such as combined loadings

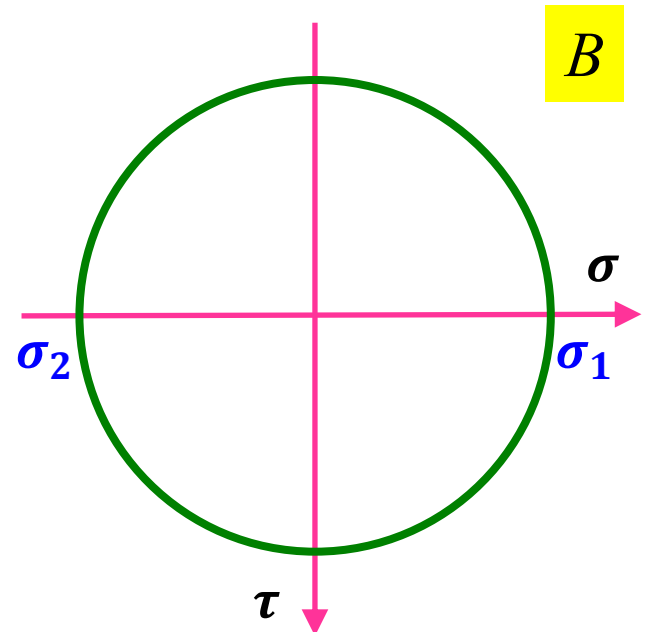
Five points (i.e., A , B , C , D , and E) are marked on the structures shown above. Sketch the Mohr's circle to represent the state of stress of each of these points.



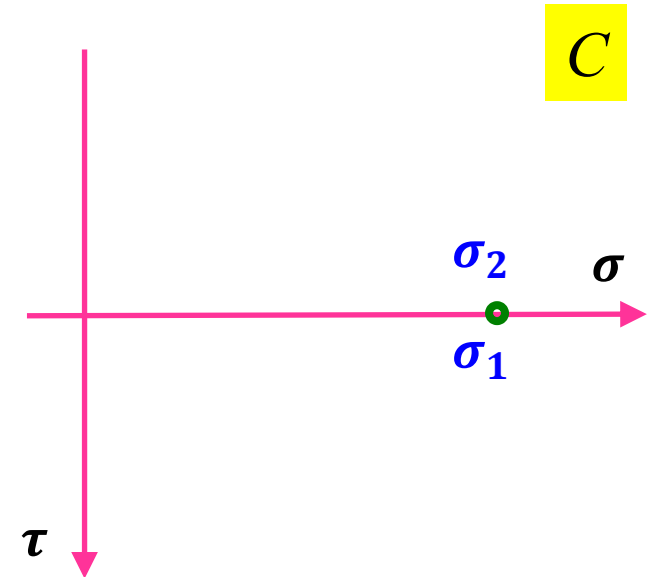
Point A: in uniaxial tension/compression, when the section is perpendicular to the axial direction of the object, there is only tensile/compressive **normal stress** acting on the plane. Therefore, the principal stresses are $\sigma_1 = P/A$, $\sigma_2 = 0$.



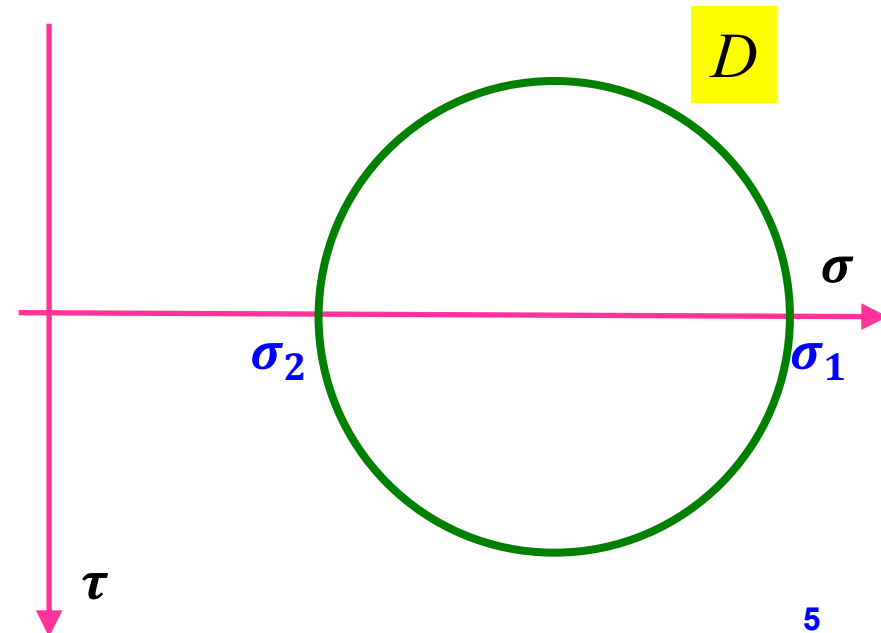
Point B: in torsion, when the section is perpendicular to the axial direction or the longitudinal direction of the shaft, there is only **shear stress** acting on the plane (**pure shear**). Therefore, the principal stresses are $\sigma_1 = T\rho/J$, $\sigma_2 = -T\rho/J$.



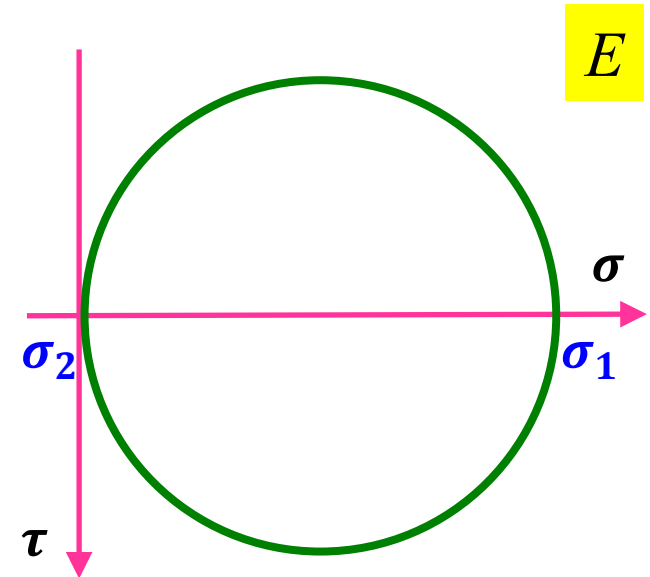
Point C: in gas filled vessels, if the section is perpendicular to the line connecting the point C and the center of the semi-sphere, there are equal **normal stresses** in two perpendicular directions (see Lecture Notes Review 5 Ch8 & Ch9, page 10). Therefore, the principal stresses are $\sigma_1 = \sigma_2 = \sigma_n$. In this case, the Mohr's circle is degenerated to a point.



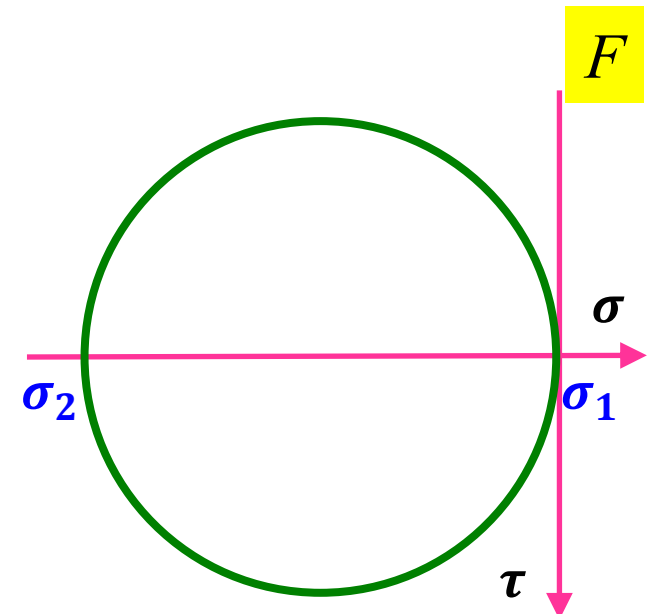
Point D: in gas filled vessels, if the section is perpendicular to the axial direction of the cylinder part, there are non-equal **normal stresses** in two perpendicular directions (one in the longitudinal direction and the other in the hoop direction; see Lecture Notes Review 5 Ch8 & Ch9, page 10). Therefore, the principal stresses are $\sigma_1 = \sigma_a$, $\sigma_2 = \sigma_n$. Note that $\sigma_a = 2\sigma_n$, thus $\sigma_1 > \sigma_2 > 0$.



Point E (top surface of the beam): in bending, when the section is perpendicular to the axial (horizontal) direction of the beam, there is only **tensile normal stress**. Therefore, the principal stresses are $\sigma_1 = \frac{Mc}{I}$, $\sigma_2 = 0$.

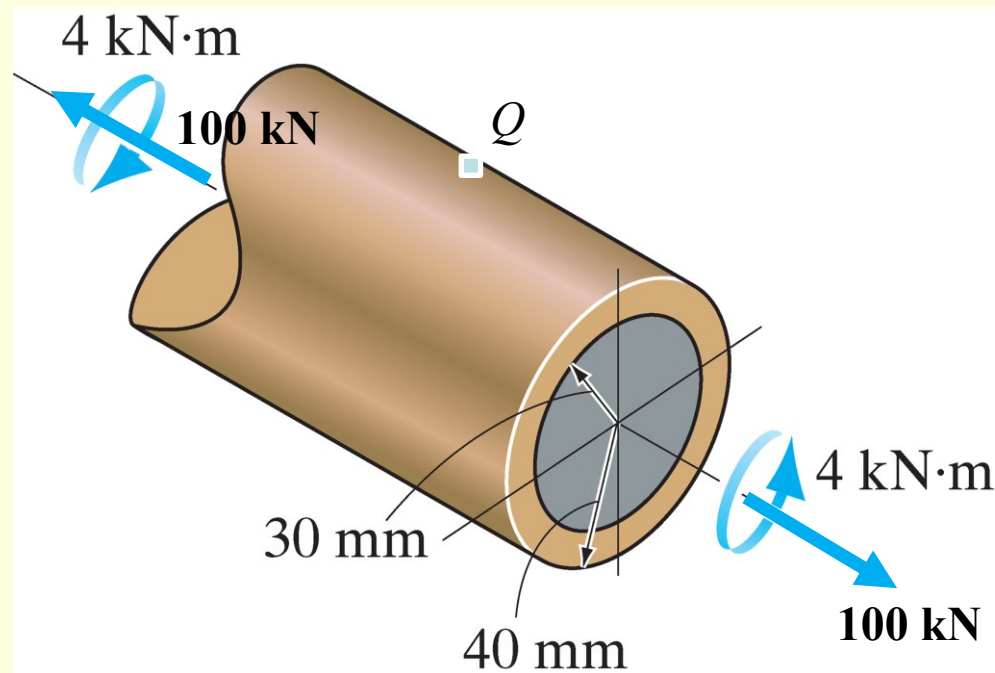


Extension: If the point is **at the bottom surface** of the beam, there is only **compressive normal stress**. In this case, the principal stresses are $\sigma_1 = 0$, $\sigma_2 = -\frac{Mc}{I}$.



IN-CLASS EXAMPLE

The hollow circular shaft is subjected to the torque and axial force shown below. Determine the principal stresses and the absolute maximum shear stress at a point Q on the surface of the shaft.



IN-CLASS EXAMPLE, cont'd

Solutions

Step 1: Stress State at the Surface Point

Both tensile force and torque will induce stress to the surface point.

Establish Oxyz coordinate system and the Q point on the **+z-axis**, with x-axis along the longitudinal direction of the shaft.

For the tensile force, the resultant tensile stress is:

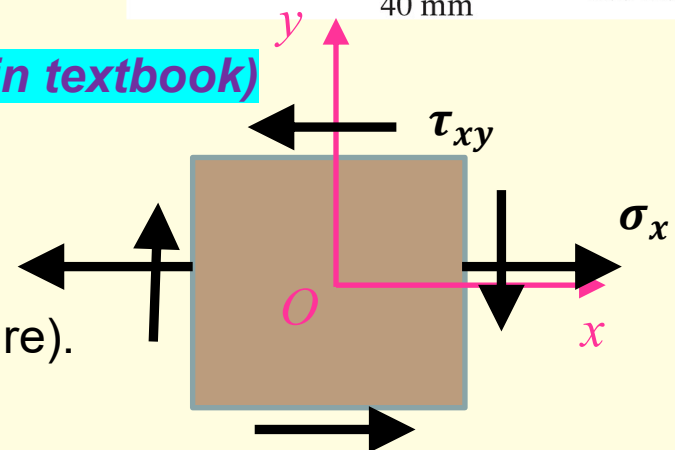
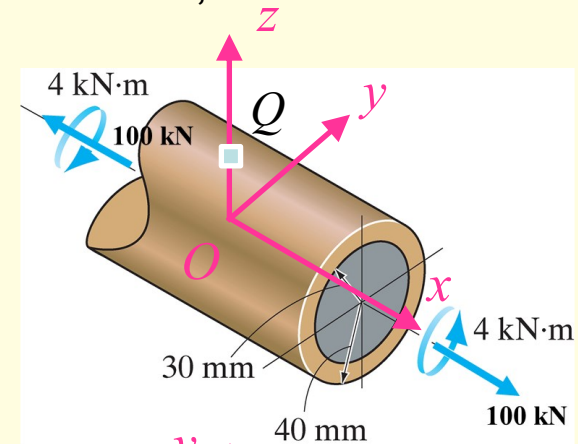
$$\sigma_x = \frac{N}{A} = \frac{100000}{\pi(0.04^2 - 0.03^2)} = 45.4 \text{ Mpa}$$

For the torque, the resultant shear stress is:

(Refer to Fig. 5-3 on p.186 & Fig. 5-6 on p.189 in textbook)

$$\tau_{xy} = -\frac{Tc}{J} = -\frac{4000 \times 0.04}{\frac{\pi}{2}(0.04^4 - 0.03^4)} = -58.2 \text{ Mpa}$$

The stress state in Oxy system is plotted (right figure).



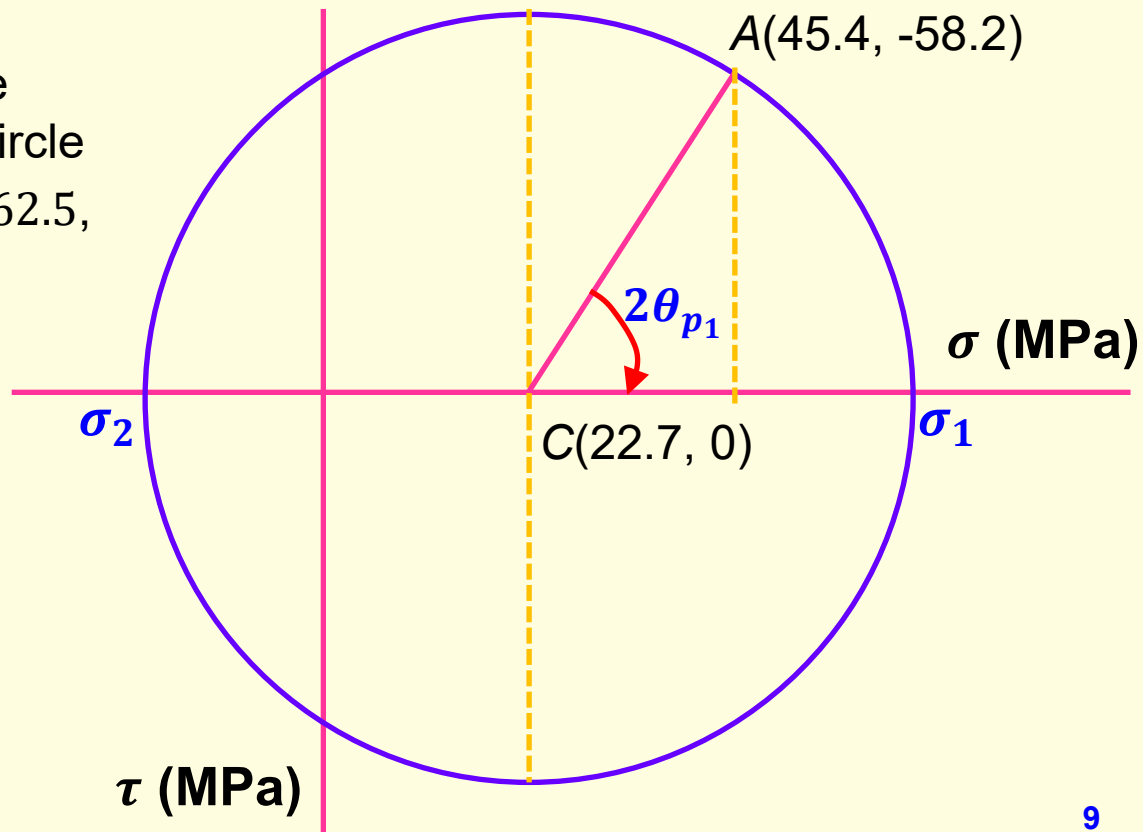
IN-CLASS EXAMPLE, cont'd

Solutions

Step 2: Construction of the Mohr's Circle

Given $\sigma_x = 45.4$ MPa, $\sigma_y = 0$, $\tau_{xy} = -58.2$ MPa, the center of the circle C is located on the σ -axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 22.7$ MPa

The point $C(22.7, 0)$ and the reference point $A(45.4, -58.2)$ are labeled. Then, the radius of the circle is $R = \sqrt{(45.4 - 22.7)^2 + 58.2^2} = 62.5$, and the circle is constructed.



IN-CLASS EXAMPLE, cont'd

Solutions, Cont'd

Step 3: Principal Stresses

The principal stresses are the two intersections of the Mohr's circle with the σ -axis

$$\sigma_1 = 22.7 + 62.5 = 85.2 \text{ MPa}$$

$$\sigma_2 = 22.7 - 62.5 = -39.8 \text{ MPa}$$

From the circle, the clockwise angle $2\theta_{p_1}$, measured from CA to the $+\sigma$ axis, is

$$2\theta_{p_1} = \tan^{-1} \left(\frac{58.2}{45.4 - 22.7} \right) = 68.7^\circ$$

Thus, $\theta = 34.35^\circ$

The clockwise rotation defines the direction of the x' -axis and σ_1 .

IN-CLASS EXAMPLE, cont'd

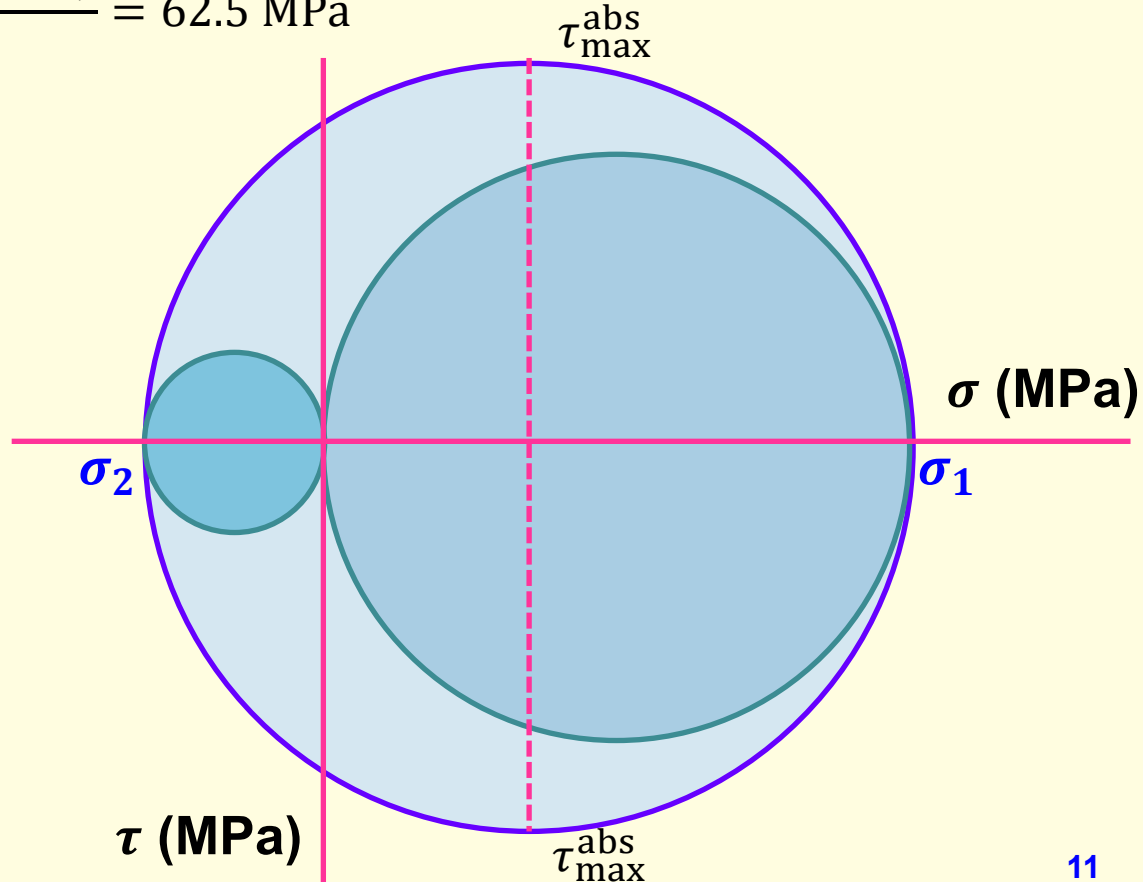
Solutions, Cont'd

Step 4: Absolute Maximum Shear Stress

Since σ_1 and σ_2 have opposite signs, applying Eq. (9-14) we have

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = R = \frac{85.2 - (-39.8)}{2} = 62.5 \text{ MPa}$$

The same results can be obtained by drawing three Mohr's circle for each orientation of an element about the x, y, and z axes, and then comparing the maximum shear stress among the three cases (right figure).



RELEVANT PRACTICE

- **Assignment 3 on Blackboard, problem #5-6 (go to Blackboard under the “Assignments” folder, **Assignment 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**
- **Optional Extra Exercise 3 on Blackboard, problem #5 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**